

Apply subtypes and domains to Vienna hiking trails

Implement and apply attribute validation techniques to geodatabase features to ensure data integrity and simplify data collection and editing.

Duration	Difficulty
1hr(s) 30mins	Intermediate

The city of Vienna, Austria, offers visitors and residents more than 446 hiking options, totaling almost 240 kilometers of hiking trails. These trails can be reached by public transportation and include access to various facilities, restaurants, and inns along the way. The city publishes and maintains a digital map of the trails, but the trail layers lack detailed attribute information, making queries and updates difficult.

A local tourism agency intends to update their website to include a web map and a downloadable app of hiking trails. To assist them, you'll update a copy of the existing trail data by implementing geodatabase attribute data validation using subtypes and domains. Then, you'll publish the updated data to the web so clients can interact with categorized trails, add and recommend new trails, and update current details such as difficulty level, duration, attractions, and accessibility.

This tutorial was last tested on May 31, 2023, using ArcGIS Pro 3.1. If you're using a different version of ArcGIS Pro, you may encounter different functionality and results.

Requirements

- ArcGIS Pro ([see options for software access](#))
- ArcGIS organizational account with a Publisher or Administrator role

Outline

[Explore layer attributes](#)

Investigate and create attribute fields.

30 minutes

[Create subtypes](#)

Create and apply a subtype.

15 minutes

[Create domains](#)

Create and apply coded value and range domains.

30 minutes

[Publish a feature service](#)

Publish a feature service with subtypes and domains applied.

15 minutes

Explore layer attributes

First, you'll investigate current hiking trail data and identify opportunities to implement and apply attribute data validation to streamline data maintenance and consistency.

Explore a map of hiking trails

You'll review line features representing hiking trails around Vienna, Austria. You'll investigate the current attribute values and consider how the use of GIS and attribute validation can support data integrity and improve organizational efficiency for users of the hiking trail data available from the City of Vienna.

1. Open the City of Vienna's [Vienna City Map](#).

This page contains the city's current hiking trail map. The trails are displayed as green lines.

Note:

Trails may not be displayed on the map by default. To turn on trails, under **Show on the map**, scroll to **City walks** and check the **City hiking trail** layer.

2. On the map, click a hiking trail and review the pop-up.



3. In the pop-up, click **Further information**.

A new page appears. It contains additional details about the trail you clicked.

While the map provides a great service, it would be even more useful if the additional details were available as attributes for each hiking trail. That way, the information could be displayed in the trail's pop-up, which would eliminate the need to navigate to a new page. Furthermore, when new trails are created, it would be a more streamlined process to populate trail attributes because geodatabase attribute validation techniques will be applied.

4. Close the additional details page and the page with the Vienna City Map.

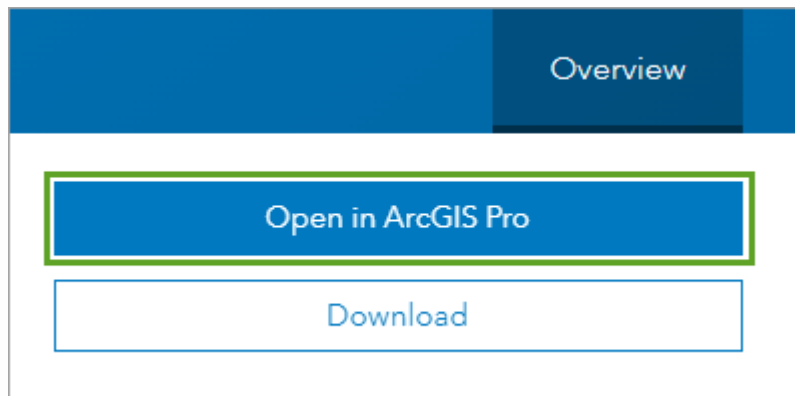
Download the data

Next, you'll download a project package containing the data used in this tutorial and open the package in ArcGIS Pro. You'll add some of the data from a geodatabase to a map.

1. Go to the [Vienna](#) item on ArcGIS Online.

This project package contains the data used in this tutorial.

2. Click **Open in ArcGIS Pro**.



The content is downloaded as a package called item.pitemx.

Note:

If you prefer, you can download and open the project package separately. To do so, click **Download**. Locate the downloaded Vienna.ppkx on your computer. If necessary, copy and move the file to a suitable location, then double-click it to open the package in ArcGIS Pro.

3. Open item.pitemx. If prompted, in ArcGIS Pro, sign in using your licensed ArcGIS account.

Note:

If you don't have access to ArcGIS Pro or an ArcGIS organizational account, [see options for software access](#).

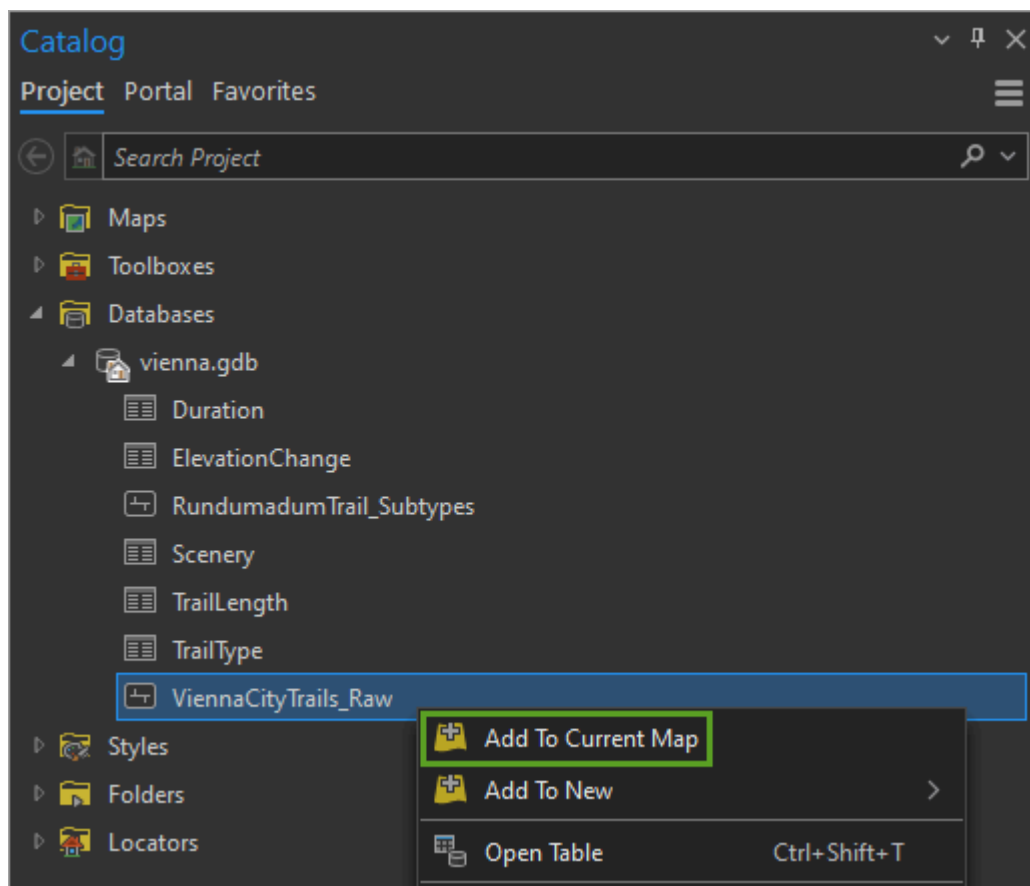
The package opens in ArcGIS Pro. It contains a map zoomed to Vienna.

4. In the **Catalog** pane, expand the **Databases** folder and expand vienna.gdb.

Tip:

If the **Catalog** pane isn't open, click the **View** tab on the ribbon. In the **Windows** group, click **Catalog Pane**.

5. Right-click **ViennaCityTrails_Raw** and choose **Add To Current Map**.



Tip:

In ArcGIS Pro, you can personalize the appearance of the user interface with either a light or a dark theme. In this tutorial, the example images use the dark theme, but you can choose the theme that you prefer. If you want to change the theme, click **Project** on the ribbon. Click **Options**. In the **Options** window, under **Application**, click **General**. Expand **Application Theme** and choose **Dark**. You'll need to restart ArcGIS Pro for your theme changes to take effect.

The **Vienna City Trails** features are displayed with default symbology.

Note:

The symbology color of the layer may differ from the example image, but it does not impact the results of the analysis.



6. In the **Contents** pane, uncheck the **Vienna City Trails** layer to turn it off.
7. In the **Catalog** pane, under **vienna.gdb**, drag **RundumadumTrail_Subtypes** onto the map. If necessary, in the **Contents** pane, turn on the **Rundumadum Trail Stages** layer.

This layer is already symbolized with a different color for each trail stage. When a layer contains subtypes, ArcGIS Pro automatically applies **Unique Values** symbology to the attribute with subtypes.

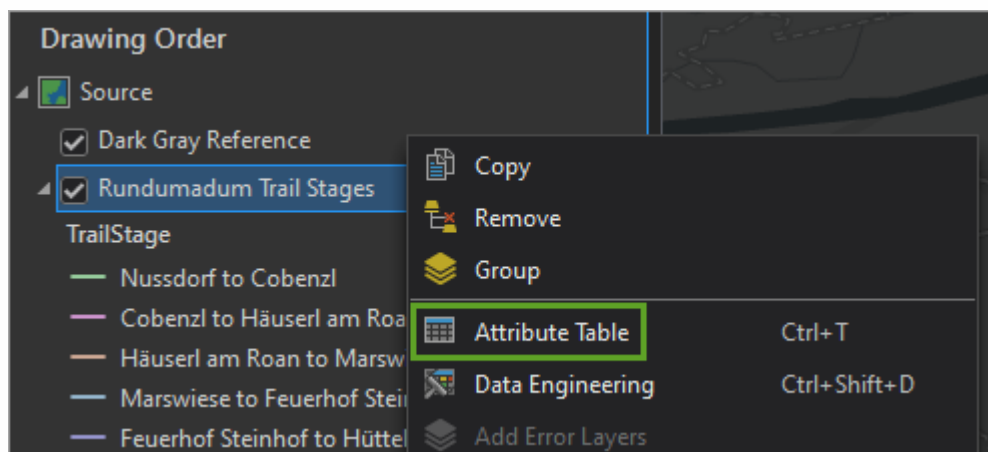


The **Rundumadum Trail Stages** layer has a subtype applied to its **TrailStage** field. Subtypes allow you to categorize or group features within a feature class and offer editors a predefined list of valid attribute choices. A feature class can have a subtype applied to only one of its attribute fields.

Explore the attributes

The **Rundumadum Trail Stages** layer is symbolized based on its attributes. To better understand the layer's attributes, you'll look at them in the layer's attribute table.

1. In the **Contents** pane, right-click the **Rundumadum Trail Stages** layer and choose **Attribute Table**.



The layer's attribute table appears. Attribute tables form the foundation of geographic data and are the fundamental building blocks of any data model. Attribute information in a table is organized into rows and columns. In ArcGIS, rows are known as records, and columns are referred to as fields.

2. If necessary, drag and dock the **Rundumadum Trail Stages** attribute table below the map.

The highlighted field, **Trail Stage Number**, represents the field to which the subtype has been applied. The field displays a description containing the trail stage name.

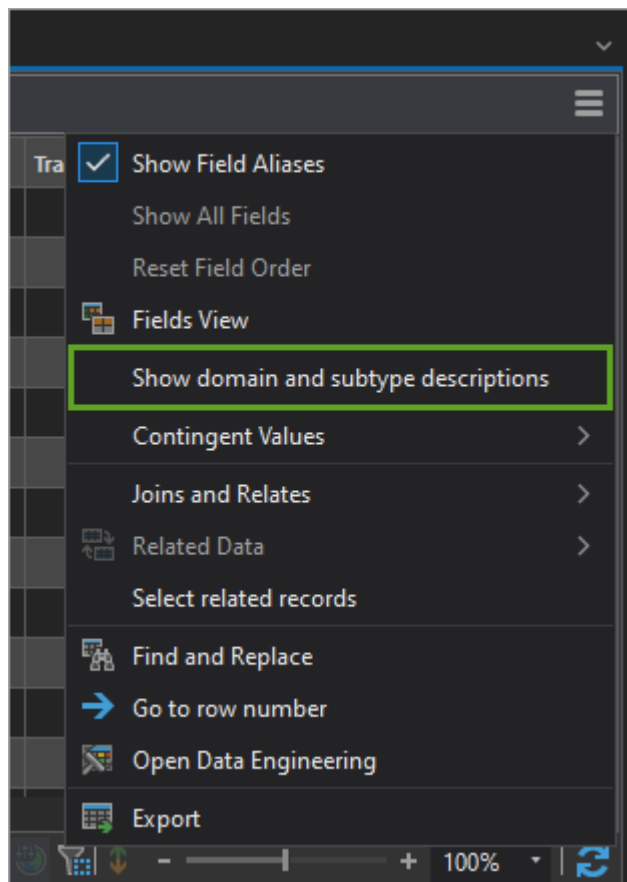
Rundumadum Trail Stages						
Field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy						
	OBJECTID_1 *	Shape *	OBJECTID	Original Name	Trail Website	Trail Stage Number
1	4	Polyline	66743	rundumadum Abschni...	http://www.wien.gv.at/l	Nationalparkcamp to Ebl
2	5	Polyline	66744	rundumadum Abschni...	http://www.wien.gv.at/l	Wienerberg to Laaer Wa
3	6	Polyline	66745	Stadtwanderweg 2; ru...	http://www.wien.gv.at/l	Cobenzl to Häuserl am R
4	7	Polyline	66746	rundumadum Abschni...	http://www.wien.gv.at/l	Hütteldorf station to Lai
5	9	Polyline	66748	rundumadum Abschni...	http://www.wien.gv.at/l	Feuerhof Steinhof to Hü
6	12	Polyline	66751	rundumadum Abschni...	http://www.wien.gv.at/l	Gerasdorf to Brünner Str
7	13	Polyline	66752	rundumadum Abschni...	http://www.wien.gv.at/l	Esslinger Furt to Himmel
8	14	Polyline	66753	rundumadum Abschni...	http://www.wien.gv.at/l	Strebersdorf to Nussdor
9	17	Polyline	66756	rundumadum Abschni...	http://www.wien.gv.at/l	Breitenleer Straße to Wa
10	26	Polyline	66784	Weitwanderweg 07; r...	http://www.wien.gv.at/l	Panozzalacke to Nationa
11	27	Polyline	66739	Weitwanderweg 04; St...	http://www.wien.gv.at/l	Nussdorf to Cobenzl
12	28	Polyline	66740	Weitwanderweg 04; V...	http://www.wien.gv.at/l	Häuserl am Roan to Mar

Note:

Another way to display a layer's attribute table is to click the layer in the **Contents** pane. On the ribbon, click the **Data** tab. In the **Table** group, click **Attribute Table**. In addition, the **Design** group contains buttons for **Fields**, **Subtypes**, and **Domains**. These are tools for working with a data model and updating and modifying the geodatabase schema.

Behind the scenes, the geodatabase stores integer values instead of descriptive names in the subtype field. The integers are paired to the descriptions via a lookup table, which results in smaller file sizes and greater efficiency.

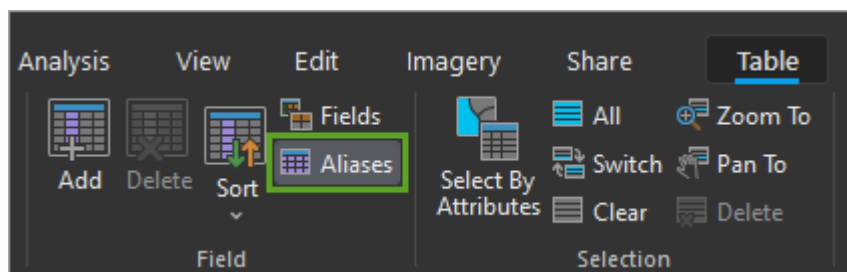
3. In the attribute table, click the options button and uncheck **Show domain and subtype descriptions**.



The table updates to show the subtype codes instead of the descriptions.

	Trail Stage Number	Trail Stage Number
t/l	16	22nd
t/l	10	10th
t/l	2	19th
t/l	6	14th
t/l	5	Fire
t/l	21	22nd
t/l	17	22nd
t/l	24	21st
t/l	19	22nd
t/l	15	22nd
t/l	1	19th
t/l	3	19th

4. Click the options button and check **Show domain and subtype descriptions** to display the subtype descriptions.
5. With the table selected, on the ribbon, click the **Table** tab. In the **Field** group, click **Aliases**.

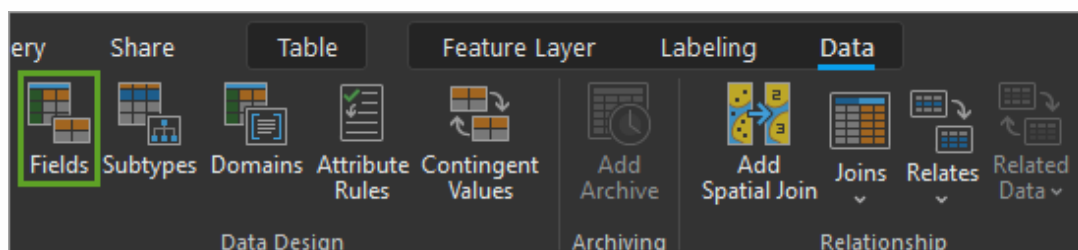


The field names switch from displaying field aliases with descriptive names to displaying the original field names. For example, the **Trail Stage Number** alias becomes the **TrailStage** field name.

OBJECTID_1 *	Shape *	OBJECTID	BEZ_TEXT	URL_INFO	TrailStage
4	Polyline	66743	rundumadum Abschni...	http://www.wien.gv.at/u	Nationalparkcamp to EBI

Aliases are a useful way to provide users with descriptive field names without compromising database field name rules that restrict the length and the use of spaces and special characters.

- On the ribbon, click **Aliases** to switch back to aliases.
- If necessary, in the **Contents** pane, click the **Rundumadum Trail Stages** layer to select it.
- On the ribbon, click the **Data** tab. In the **Data Design** group, click **Fields**.



The **Fields** view opens. It displays a layer's attribute fields and properties. In this view, you can create fields, delete fields, and modify existing fields.

The ***TrailStage** field name is bold and preceded by an asterisk to denote that it has a subtype applied. The field's data type is set to **Short**, which means short integer. Fields with a subtype must be a short or long integer field. The field has an alias of **Trail Stage Number**.

Rundumadum Trail Stages

Fields: Rundumadum Trail Stages

Current Layer

Rundumadum Trail Stages

	☒ Visible	☐ Read Only	Field Name	Alias	Data Type	☒ Allow NULL
	☒	☒	OBJECTID_1	OBJECTID_1	Object ID	☐
	☒	☐	Shape	Shape	Geometry	☒
	☒	☐	OBJECTID	OBJECTID	Double	☒
	☒	☐	BEZ_TEXT	Original Name	Text	☒
	☒	☐	URL_INFO	Trail Website	Text	☒
	☒	☐	*TrailStage	Trail Stage Number	Short	☒
	☒	☐	TrailStart	Trail Start	Text	☒
	☒	☐	TrailEnd	Trail End	Text	☒

One of the fields is called **BEZ_TEXT**. Because the field name is not descriptive, the alias **Original Name** was set for this field.

9. For **BEZ_TEXT**, double-click the cell containing the alias **Original Name**.

The cell is highlighted, indicating that you can optionally update or change the current alias. You won't make a change at this time.

10. Deselect the cell by clicking elsewhere in the table.

The **Data Type** column contains the data type for each field. In this table, fields are primarily set to **Text**, **Float**, and **Short**. Internal fields are set to **Double**. Trail locations (coordinates) are stored in a special field type called **Geometry**.

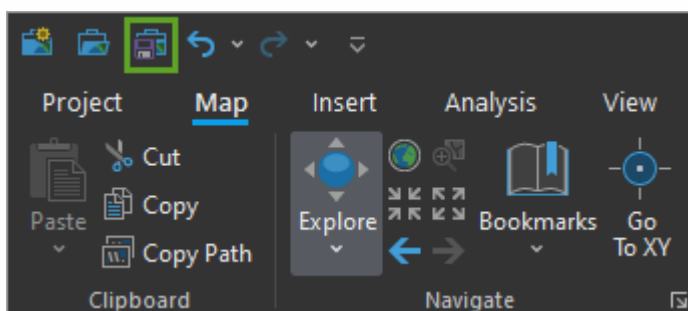
Alias	Data Type	☒ Allow NULL
OBJECTID_1	Object ID	☐
Shape	Geometry	☒
OBJECTID	Double	☒
Original Name	Text	☒
Trail Website	Text	☒
Trail Stage Number	Short	☒
Trail Start	Text	☒
Trail End	Text	☒
Trail Length	Float	☒
Estimated Time	Text	☒

Each field in a table stores a specific type of data, as shown in the following table:

Field type	Description
Text	String of characters
Float	Decimal numbers between -3.4E38 and 1.2E38
Double	Decimal numbers between -2.2E308 and 1.8E308
Short	Whole numbers between -32,768 and 32,767
Long	Whole numbers between -2,147,483,648 and 2,147,483,647
Date	Date or time or both date and time
Blob	A collection of binary data such as images, audio, and other multimedia

Field type	Description
Raster	Raster images
GUID	Globally unique identifier
Geometry	Stores x,, coordinate pairs representing spatial features

11. Close the **Fields** view and **Rundumadum Trail Stages** attribute table. In the **Contents** pane, uncheck the **Rundumadum Trail Stages** layer.
12. On the **Quick Access Toolbar**, click the **Save** button.



The project is saved.

Add a new field

Next, you'll add a new short integer field that will be used to create a subtype for the **Vienna City Trails** layer.

1. In the **Contents** pane, turn on the **Vienna City Trails** layer. Right-click the layer, point to **Data Design**, and choose **Fields**.

The **Fields** view for the layer appears.

2. If necessary, position and dock the pane under the map.
3. Scroll to the bottom of the **Fields** view and click **Click here to add a new field**.

☒	☐	TrailCode	Trail Code	Short
☒	☒	Shape_Length	Shape_Length	Double
Click here to add a new field.				

Tip:

New fields can also be added from the ribbon on the **Fields** tab in the **Changes** group.

4. In the new row, double-click the empty cell in the **Field Name** column. Type TrailRating.

Note:

As you make edits, a green indicator appears next to edited rows. A red indicator next to a row indicates an error that will prevent that row from being committed when **Save** is clicked. To read the

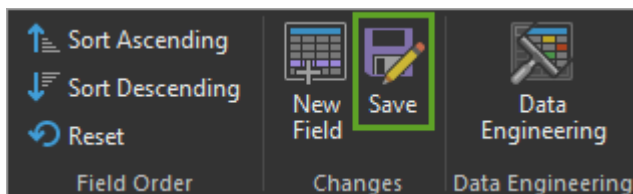
error, point to the red indicator. When you correct the error, the indicator turns green again.

- Double-click the empty cell for **Alias** and type Trail Rating.
- Double-click the **Data Type** cell and choose **Short**.
- Check the **Highlight** box.

☒ Visible	☐ Read Only	Field Name	Alias	Data Type	☒ Allow NULL	☐ Highlight
☒	☐	Elevation	Average Trail Elevation	Short	☒	☐
☒	☐	TrailType	Type of trail	Text	☒	☐
☒	☐	TrailCode	Trail Code	Short	☒	☐
☒	☒	Shape_Length	Shape_Length	Double	☒	☐
☒	☐	TrailRating	Trail Rating	Short	☒	☒
Click here to add a new field.						

The **Number Format** property allows you to change the display of numeric values, which is useful if you want to limit the number of decimal places shown or express numbers as a percent. However, this won't be necessary for your new **Trail Rating** field. Even though it will store numeric values, it will not display them as numbers. The integer values will be substituted with subtype descriptions instead.

- On the ribbon, on the **Fields** tab, in the **Changes** group, click **Save**.



Note:

When you save changes made while working in the **Fields** view, you are updating the geodatabase structure, also known as the geodatabase schema. This is different than editing features in a feature class or attribute table, where you are only changing content.

- Close the **Fields** view.
- In the **Contents** pane, right-click the **Vienna City Trails** layer and choose **Attribute Table**.

The **Trail Rating** field has been added to the end of the attribute table.

Type of trail	Trail Code	Shape_Length	Trail Rating
< Null >	1	0.093314	< Null >
< Null >	6	0.156438	< Null >
< Null >	4	0.06343	< Null >
< Null >	4	0.108996	< Null >
< Null >	8	0.101569	< Null >
< Null >	4	0.009476	< Null >
< Null >	5	0.118171	< Null >

Populate the field

By default, new fields have no values. You'll edit the **Trail Rating** field you created to include values based on the length of time it takes to complete each trail.

1. In the attribute table, right-click the **Estimated Time** field and choose **Sort Ascending**.

Trail End	Estimated Time	Facilities	Route Description
Nußdorfer Platz	3 to 4 hours		ussdorf Square - Do...
tram 60 terminus Roda...	4 to 4.5 hours		am 60 Rodaun - Ketz...
Tram 49 Station Rettic...	2.5 to 3 hours		am 49 Rettichgasse...
U3 station Ottakring	2.5 to 3 hours		station Ottakring -...
Postbus 450 Station K...	3 to 4 hours		stbus 450 Station K...
Tram 49 Station Rettic...	2.5 to 3 hours		am 49 to Rettichgass...
tram 31 terminus Stam...	3 to 4 hours		am 31 Endstation St...

For the trails, the rating is based on the estimated time to complete the trail. For example, trails that can be completed in 2.5 to 3 hours are rated as easy.

2. Drag the **Estimated Time** field next to the **Trail Rating** field.
3. For the **Trail Rating** field, double-click the first cell, type 1, and press Enter.

Shape_Length	Estimated Time	Trail Rating
0.06343	2.5 to 3 hours	1
0.108996	2.5 to 3 hours	< Null >
0.009476	2.5 to 3 hours	< Null >
0.093314	3 to 4 hours	< Null >
0.101569	3 to 4 hours	< Null >

4. In the second cell, type 1 and press Enter.
5. In the **Save Edits** window, click **Yes**.
6. In the attribute table, continue updating the **Trail Rating** field values using the following table:

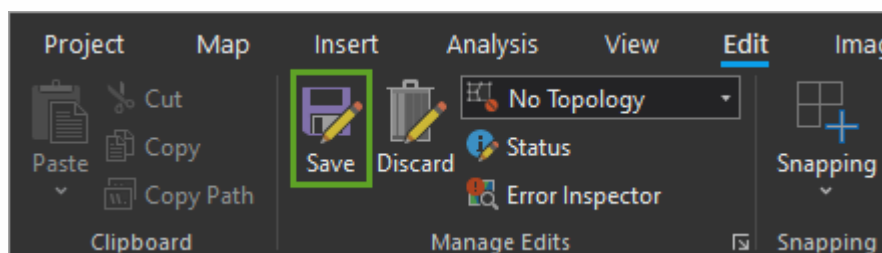
Estimated Time	Trail Rating
2.5 to 3 hours	1
3 to 4 hours	2
3.5 to 4 hours	2
4 to 4.5 hours	3

Estimated Time	Trail Rating
4 to 5 hours	3

7. In the **Vienna City Trails** attribute table, confirm that you successfully updated all **Trail Rating** field values.

Estimated Time	Trail Rating
2.5 to 3 hours	1
2.5 to 3 hours	1
2.5 to 3 hours	1
3 to 4 hours	2
3 to 4 hours	2
3 to 4 hours	2
3 to 4 hours	2
3 to 4 hours	2
3 to 4 hours	2
3 to 4 hours	2
3.5 to 4 hours	2
4 to 4.5 hours	3
4 to 5 hours	3

8. On the ribbon, on the **Edit** tab, in the **Manage Edits** group, click **Save**.



9. In the **Save Edits** window, click **Yes**.
10. Close the attribute table. Save the project.

You've successfully added and populated the field in preparation for applying a subtype based on trail rating. Next, you'll add the subtype descriptions and review the results.

Create subtypes

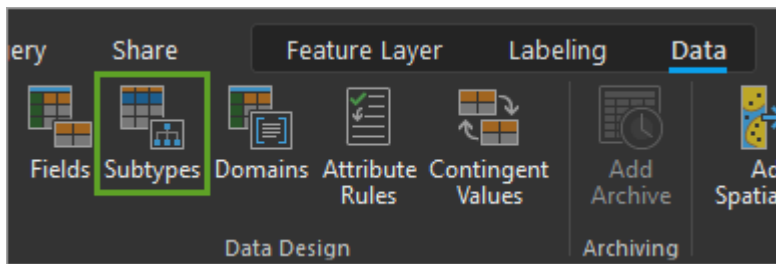
Next, you'll create a subtype on a field and update codes and descriptions.

Create subtypes for trail ratings

Subtypes allow you to categorize a layer into groups of features that share the same attribute. Using the **Subtypes** view, you can review the subtypes associated with layers, make edits to the properties of the subtypes, or create subtypes for a layer.

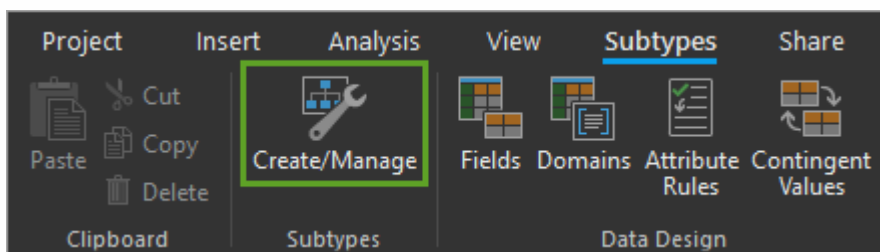
The current **Vienna City Trails** layer does not have any subtypes defined. You'll create a subtype for the **Trail Rating** field you added and populated.

1. In the **Contents** pane, click the **Vienna City Trails** layer to select it.
2. On the ribbon, click the **Data** tab. In the **Data Design** group, click **Subtypes**.



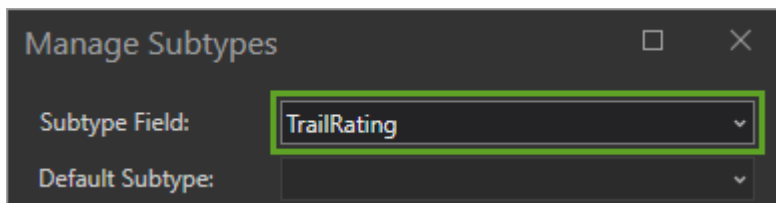
The **Subtypes** view appears. It lists all of the layer's fields in a table.

3. If necessary, resize the **Subtypes** view to display the **TrailRating** field.
4. On the ribbon, in the **Subtypes** tab, in the **Subtypes** group, click **Create/Manage**.



The **Manage Subtypes** window appears.

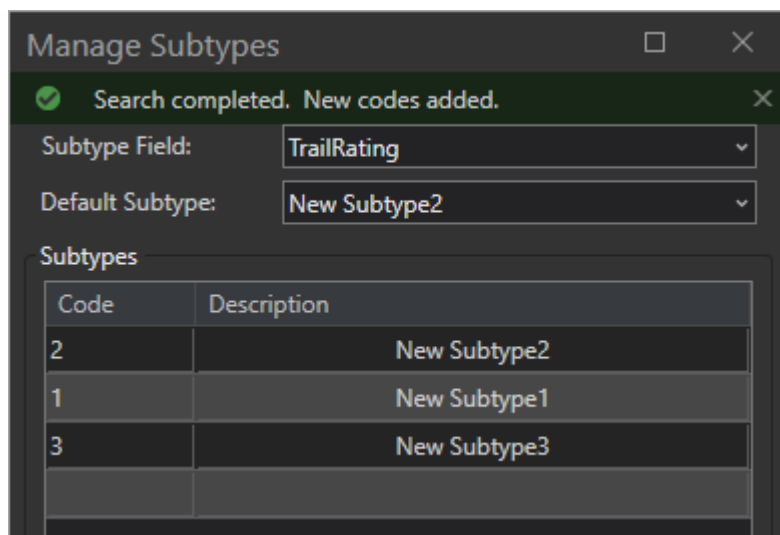
5. In the **Manage Subtypes** window, for **Subtype Field**, choose **TrailRating**.



You could add a trail rating code and a description for each subtype you need to generate. An easier way to populate the subtype codes is to allow the tool to discover each unique code in the specified subtype field. This is useful if you have many subtypes to create.

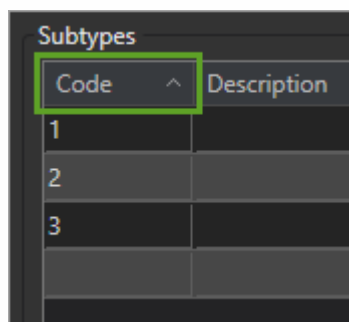
6. In the **Manage Subtypes** window, click **Discover codes**.

The **Manage Subtypes** pane updates to display three subtype codes. These represent the **Trail Rating** field values you assigned to the various trails based on the **Estimated Time** field.



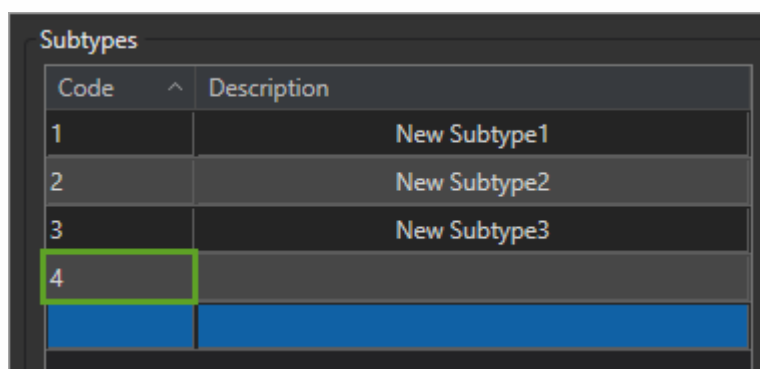
Next, you'll sort the codes and provide a description for each subtype.

- Click **Code** to sort the subtype codes in ascending order.



You'll add a subtype for new trails that do not yet have a trail rating assigned.

- Double-click the empty cell below **3**, type 4, and press Enter.



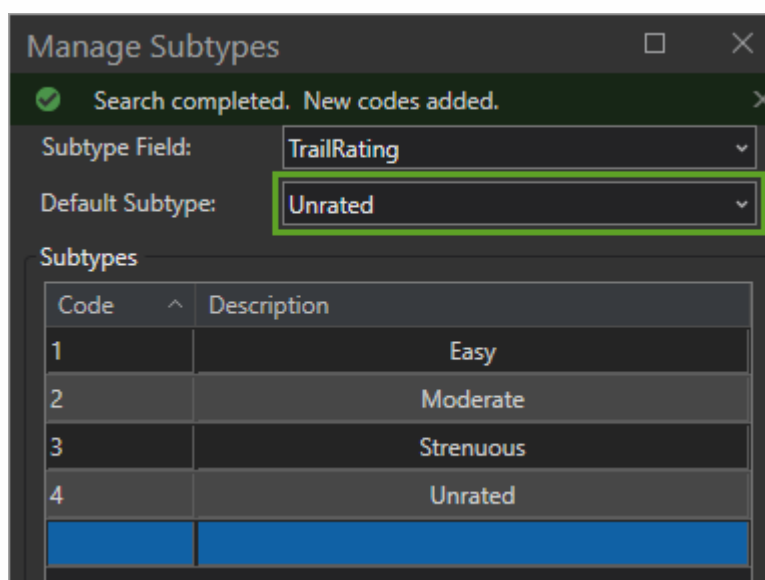
Next, you'll provide a description for each subtype.

- Under **Description**, double-click New Subtype1. Type Easy and press Enter.
- Provide a description for each additional subtype, using the following table:

Trail rating	Subtype code
1	Easy

Trail rating	Subtype code
2	Moderate
3	Strenuous
4	Unrated

11. For **Default Subtype**, choose **Unrated**.



12. Click **OK**.

The subtype is created and assigned. The **Manage Subtypes** window closes.

13. On the ribbon, on the **Subtypes** tab, in the **Changes** group, click **Save**.

In the **Subtypes** view, **TrailRating** is identified as the subtype field due to it being bold and preceded by an asterisk. In addition, each subtype has the default value set to the subtype code. Columns for default values and domains are added to each subtype.

Subtypes: Vienna City Trails									
Subtype Name >		Easy		Moderate		Strenuous		Unrated	
Field Name	Data Type	Domain	Default Value	Domain	Default Value	Domain	Default Value	Domain	Default Value
TrailType	Text								
TrailCode	Short								
Shape_Length	Double								
*TrailRating	Short	1		2		3		4	

14. Close the **Subtypes** view. In the **Contents** pane, right-click **Vienna City Trails** and choose **Attribute Table**.

In the attribute table, the **Trail Rating** field is updated to display the subtype descriptions, rather than the numeric values.

Estimated Time	Trail Rating
2.5 to 3 hours	Easy
2.5 to 3 hours	Easy
2.5 to 3 hours	Easy
3 to 4 hours	Moderate
3 to 4 hours	Moderate

15. Close the attribute table.

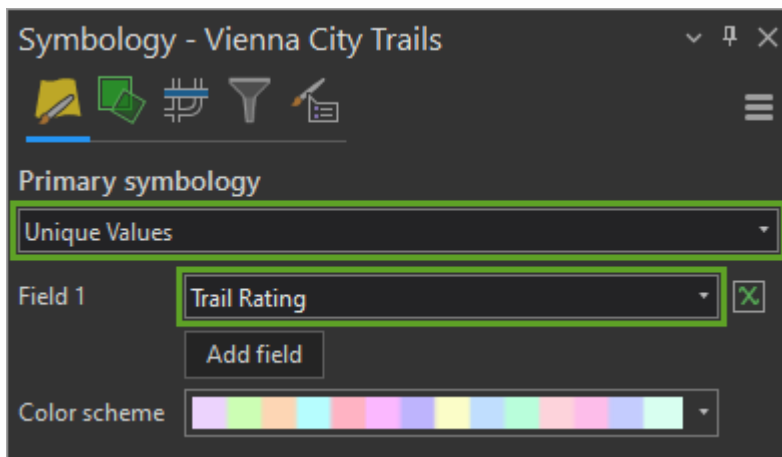
Symbolize the trails

Next, you'll symbolize the trails layer using the subtype field.

1. In the **Contents** pane, right-click the **Vienna City Trails** layer and choose **Symbology**.

The **Symbology** pane appears.

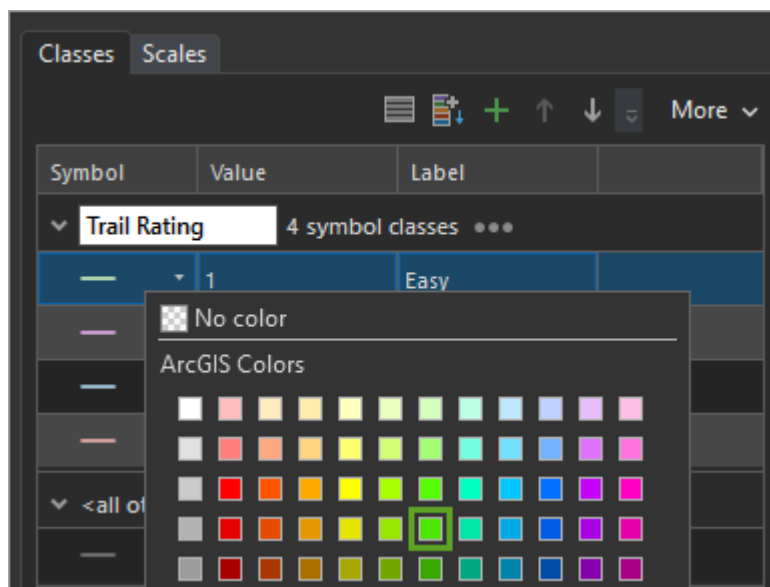
2. In the **Symbology** pane, for **Primary symbology**, choose **Unique Values**. For **Field 1**, choose **Trail Rating**.



3. In the **Classes** grid, right-click the line symbol associated with **Easy** trails. In the color picker, choose **Quetzel Green**.

Tip:

Point to a color to see its name.



Tip:

In the color picker, you can see the name of a color by pointing to it.

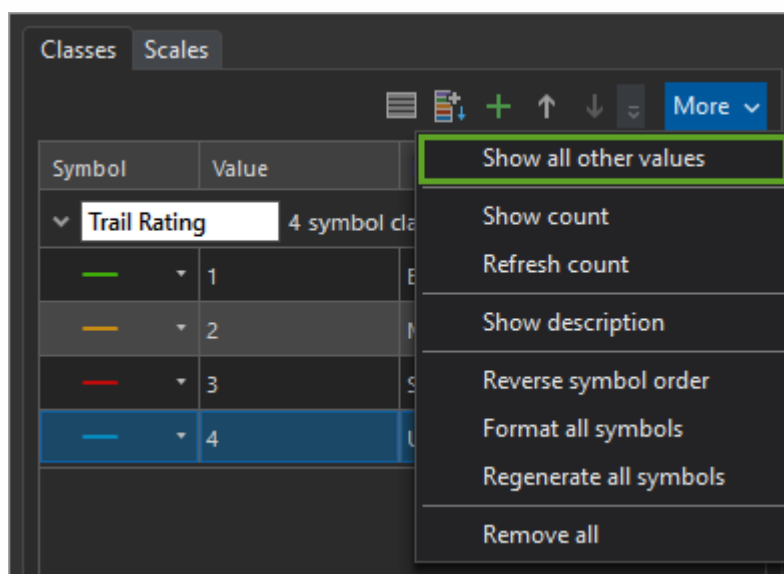
The easy trails on the map are now green.

4. Change the colors of the other symbols as follows:

- For **Moderate**, choose **Electron Gold** (fourth column, third row of the color picker).
- For **Strenuous**, choose **Mars Red** (second column, third row of the color picker).
- For **Unrated**, choose **Moorea Blue** (ninth column, fourth row of the color picker).

There is an additional symbol class that you do not need to show. Because you've already created a subtype for unrated trails, you can remove the **all other values** symbol class.

5. Click **More** and uncheck **Show all other values**.



Your map is now symbolized to more intuitively reflect trail ratings.



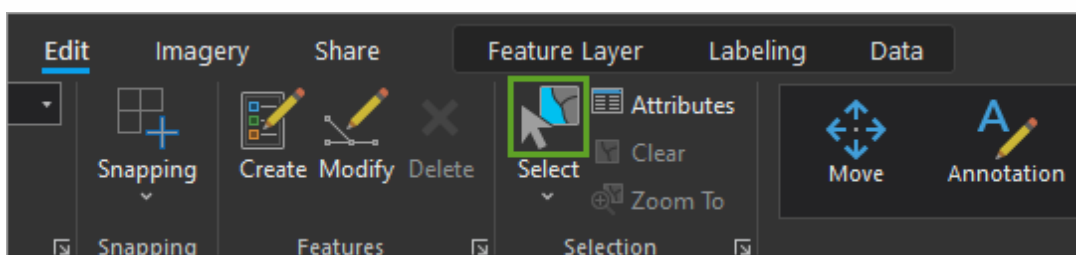
6. Close the **Symbology** pane. Save the project.

Use a subtype

You successfully updated the **Vienna City Trails** feature class by applying a subtype field on the city trails. Subtypes not only enforce integrity by allowing you to choose from a list of valid values, they also allow you to categorize your data. You can set default values on fields in each of the subtypes that will automatically apply when you create features.

Next, you'll add domains to a geodatabase and apply them to individual fields of existing feature classes.

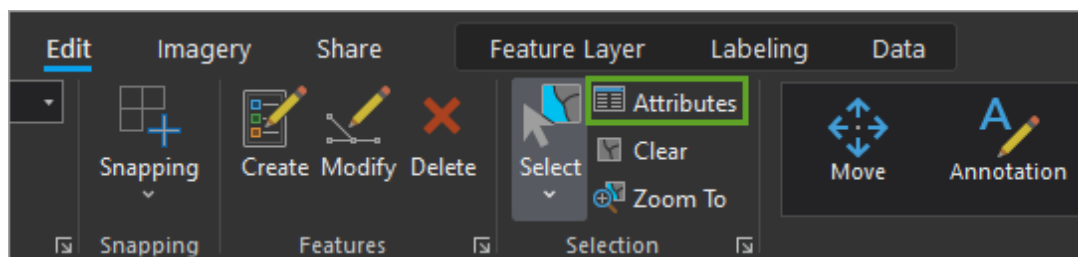
1. In the **Contents** pane, click the **Vienna City Trails** layer to select it.
2. On the ribbon, click the **Edit** tab. In the **Selection** group, click the **Select** button.



3. On the map, click any trail feature.

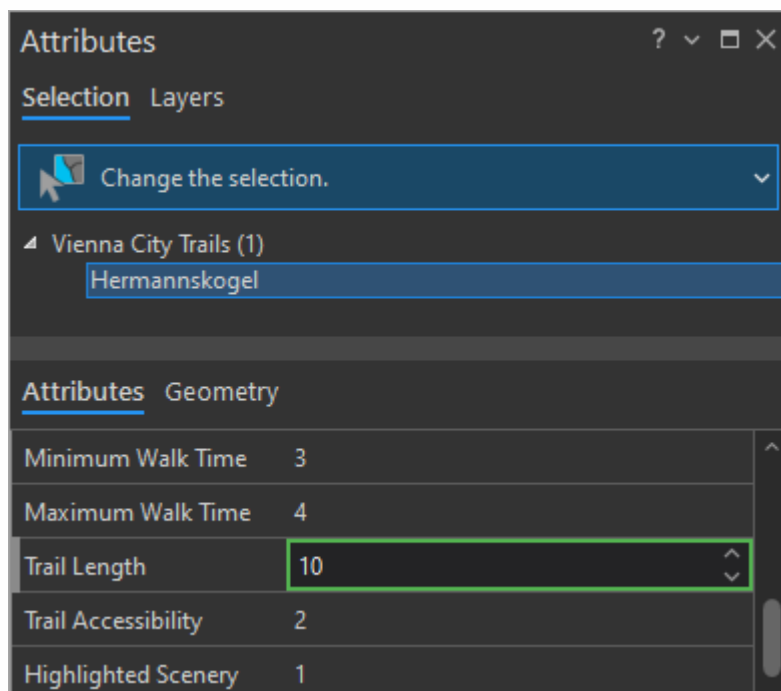
The trail you click is selected, causing it to be highlighted blue.

4. On the **Edit** tab, in the **Selection** group, click **Attributes**.



The **Attributes** pane appears. In this pane, you can edit the attribute field values for the selected trail.

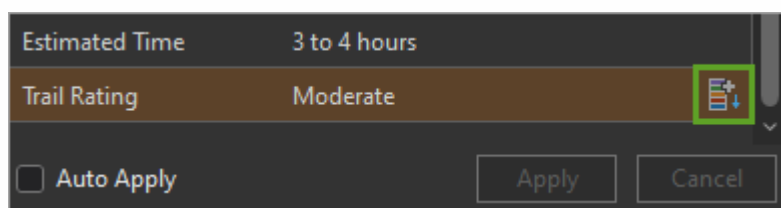
5. In the **Attributes** pane, click the value for **Trail Length** field.



The field highlights, indicating that you can update its value by typing a new value. Currently, you can update the **Trail Length** field value with any value. There is no mechanism to verify or limit what values are added. You will address this problem later.

You will not make a change to the **Trail Length** field value at this time.

6. Scroll to the bottom of the **Attributes** pane. For the **Trail Rating** field, click the **Choose Symbol Class** button.



A list of descriptions appears, allowing you update the current subtype.

You cannot type a custom **Trail Rating** value. Instead, you must select one of the valid values from the list, maintaining data consistency. This functionality is a major reason for using subtypes.

You won't change the **Trail Rating** field value at this time.

7. Close the **Attributes** pane.

- On the **Edit** tab, in the **Selection** group, click **Clear**.
- Save the project.

You've successfully updated the **Vienna City Trails** feature class in the geodatabase by applying a subtype field to the city trails.

Create domains

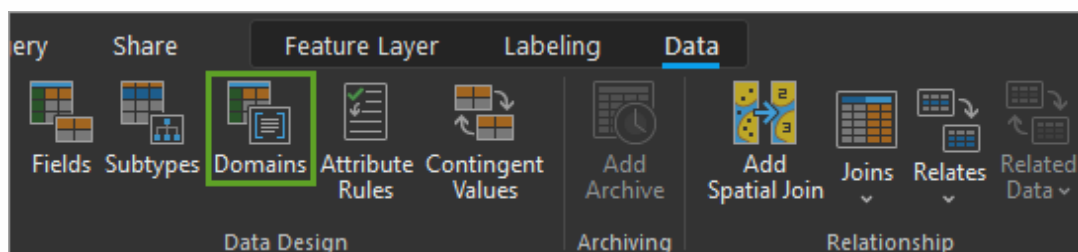
Next, you'll add domains to a geodatabase and apply them to individual fields of existing feature classes.

Create domains using a view

Domains are similar to subtypes in that they enforce data integrity by offering a list of valid values to choose from while populating attribute fields. Unlike subtypes, however, domains are used to regulate the values permitted in any particular attribute field for a table, feature class, or subtype.

Domains are stored in a geodatabase and can be applied to multiple feature classes and tables within the geodatabase. Depending on your requirements, attribute domains are applied either as range domains or coded value domains. Range domains define a field's minimum and maximum values. They can be used in short, long, float, double, and date attribute fields. Coded value domains specify a valid set of values for an attribute, giving both the allowed actual value and a more easily interpreted description of what the code represents.

- In the **Contents** pane, select **Vienna City Trails**.
- On the ribbon, click the **Data** tab. In the **Data Design** group, click **Domains**.



The **Domains** view appears. In this view, you can review existing domains, edit their properties and values, and create domains.

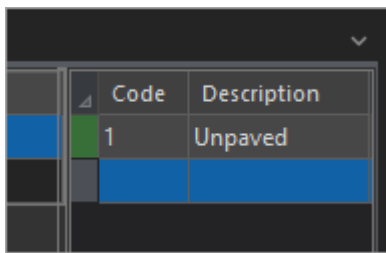
Next, you'll create a domain.

- In the **Domains** view, double-click the empty cell for **Domain Name** and type Accessibility. For **Description**, type Trail accessibility.
- For **Field Type**, choose **Short**. For **Domain Type**, confirm that **Coded Value Domain** is selected.

Source				
*Domains: vienna				
Domain Name	Description	Field Type	Domain Type	Split Policy
Accessibility	Trail accessibility	Short	Coded Value Domain	Default

When the domain type is a coded value domain, an additional table at the end of the row is used to input domain codes and descriptions.

5. For **Code**, type 1. For **Description**, type Unpaved and press Enter.



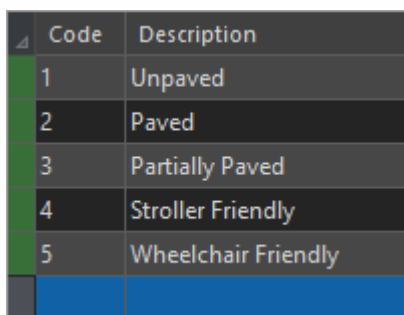
Code	Description
1	Unpaved

Once you add one code and description, another row is added.

6. Add **Code** and **Description** values using the following table:

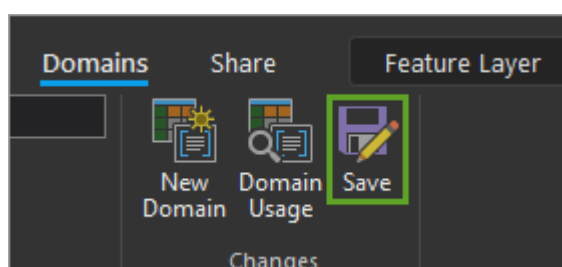
Code	Description
1	Unpaved
2	Paved
3	Partially Paved
4	Stroller Friendly
5	Wheelchair Friendly

There are now five codes and descriptions.



Code	Description
1	Unpaved
2	Paved
3	Partially Paved
4	Stroller Friendly
5	Wheelchair Friendly

7. On the ribbon, on the **Domains** tab, in the **Changes** group, click **Save**.

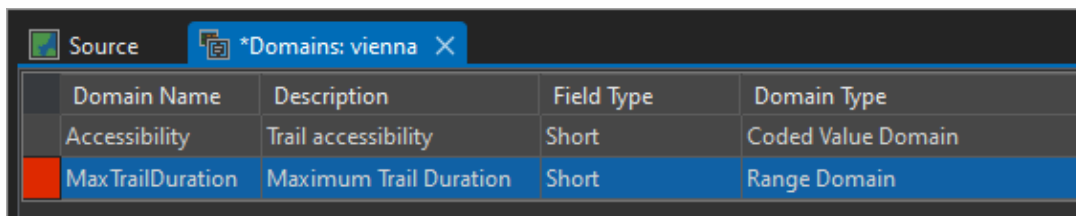


8. On the **Domains** tab, in the **Changes** group, click **New Domain**.

A new domain is added to the view.

9. For the new domain, enter the following information:

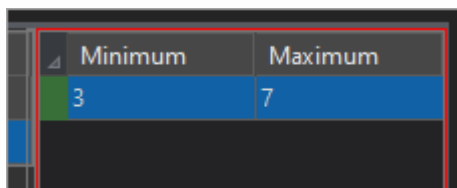
- For **Domain Name**, type MaxTrailDuration.
- For **Description** type Maximum Trail Duration.
- For **Field Type**, choose **Short**.
- For **Domain Type**, choose **Range Domain**.



Domain Name	Description	Field Type	Domain Type
Accessibility	Trail accessibility	Short	Coded Value Domain
MaxTrailDuration	Maximum Trail Duration	Short	Range Domain

When a range domain is selected, the additional table provides for the input of minimum and maximum values to limit or constrain the numerical values that can be used in the field.

10. For **Minimum**, type 3. For **Maximum**, type 7.



Minimum	Maximum
3	7

11. On the ribbon, on the **Domains** tab, in the **Changes** group, click **Save**.

12. Create a range domain for minimum trail duration using the following parameters:

- For **Domain Name**, type MinTrailDuration.
- For **Description**, type Minimum Trail Duration.
- For **Field Type**, choose **Short**.
- For **Domain Type**, choose **Range Domain**.
- For **Minimum Value**, type 1.
- For **Maximum Value**, type 2.

13. On the ribbon, on the **Domains** tab, in the **Changes** group, click **Save**.

The **Domains** view now has three domains.

14. Add three more range domains using the following parameters:

Domain Name	Description	Field Type	Domain Type	Minimum Value	Maximum Value
ShortTrailLength	Average kilometer length of a short trail	Short	Range Domain	2	9

Domain Name	Description	Field Type	Domain Type	Minimum Value	Maximum Value
MediumTrailLength	Average kilometer length of a medium trail	Short	Range Domain	10	13
LongTrailLength	Average kilometer length of a long trail	Short	Range Domain	14	15

You've created a total of six domains.

Domain Name	Description	Field Type	Domain Type
Accessibility	Trail accessibility	Short	Coded Value Domain
MaxTrailDuration	Maximum Trail Duration	Short	Range Domain
MinTrailDuration	Minimum Trail Duration	Short	Range Domain
ShortTrailLength	Average kilometer length of a short trail	Short	Range Domain
MediumTrailLength	Average kilometer length of a medium trail	Short	Range Domain
LongTrailLength	Average kilometer length of a long trail	Short	Range Domain

15. On the ribbon, on the **Domains** tab, in the **Changes** group, click **Save**.

16. Close the **Domains** view.

Create domains using a tool

Creating coded value domains with many codes and descriptions can be a time consuming and laborious process. To speed up the process, you'll use a geoprocessing tool to create additional coded value domains from existing tables in the geodatabase.

The **Table To Domain** geoprocessing tool is used to create or update a coded value domain with values from a table.

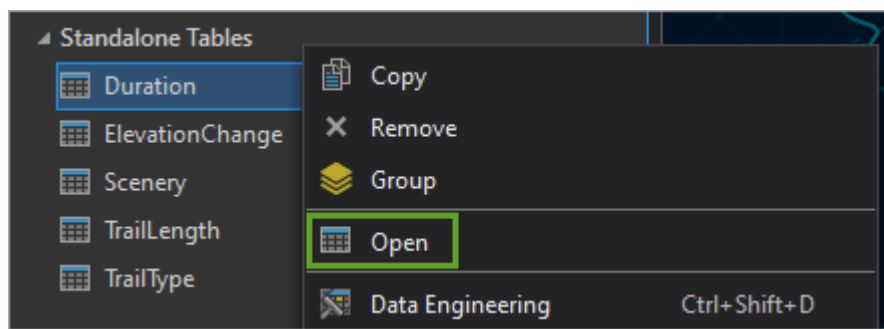
Note:

Attribute domains are a property of the geodatabase and can be shared across feature classes, tables, and subtypes in a geodatabase. They provide an effective way of enforcing data integrity by limiting what can be inserted into a field using the codes in a coded value domain or the minimum or maximum range in a range domain.

1. In the **Catalog** pane, expand the **Maps** folder. Right-click **ReferenceMap** and choose **Open**.

A map view appears. This map contains several standalone tables.

2. In the **Contents** pane, right-click the **Duration** table and choose **Open**.

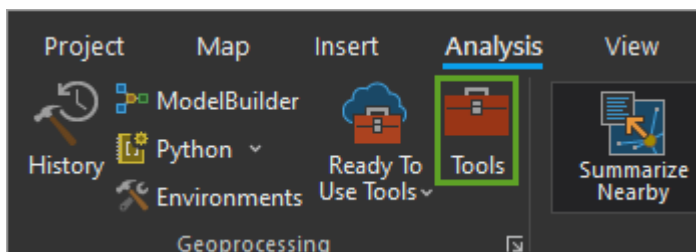


The table contains values that can be used in a **Trail Duration** domain.

Duration			
Field: Add Calculate Selection: Select			
	OID *	TrailDuration	TrailDurationTimes
1	1	1	< 20 min
2	2	2	< 1 hour
3	3	3	1 – 2 hours
4	4	4	3 – 4 hours
5	5	5	> 5 hours
Click to add new row.			

Next, you'll use a geoprocessing tool to create a domain and assign values to it from this table.

3. Close the **Duration** table and the **ReferenceMap** map.
4. On the ribbon, on the **Analysis** tab, in the **Geoprocessing** group, click **Tools**.



The **Geoprocessing** pane appears.

5. In the **Geoprocessing** pane, search for and open the **Table To Domain** tool.
6. In the **Table To Domain** pane, set the following parameters:
 - For **Input Table**, choose **Duration**.
 - For **Code Field**, choose **TrailDuration**.
 - For **Description Field**, choose **TrailDurationTimes**.
 - For **Input Workspace**, browse to the project's **Databases** folder and choose **vienna.gdb**.
 - For **Domain Name**, accept the default name **TrailDuration**.
 - For **Domain Description**, type Time needed to complete trail.

Geoprocessing ⌵ 📌 ✕

⬅ **Table To Domain** ➕

Parameters **Environments** ?

Input Table
Duration 📁

Code Field
TrailDuration ⌵

Description Field
TrailDurationTimes ⌵

Input Workspace
vienna.gdb 📁

Domain Name
TrailDuration

Domain Description
Time needed to complete trail

Update Option
Append the values ⌵

7. Click **Run**.

The tool runs. When it finishes, no changes are made to the map. You can confirm that the tool ran successfully by checking the **Domains** view.

8. In the **Contents** pane, select **Vienna City Trails**. On the ribbon, click the **Data** tab. In the **Data Design** group, click **Domains**.

The **Domains** view appears. The **TrailDuration** domain has been successfully added.

Domain Name	Description	Field Type	Domain Type
Accessibility	Trail accessibility	Short	Coded Value Domain
LongTrailLength	Average kilometer length of a long trail	Short	Range Domain
MaxTrailDuration	Maximum Trail Duration	Short	Range Domain
MediumTrailLength	Average kilometer length of a medium trail	Short	Range Domain
MinTrailDuration	Minimum Trail Duration	Short	Range Domain
ShortTrailLength	Average kilometer length of a short trail	Short	Range Domain
TrailDuration	Time needed to complete trail	Short	Coded Value Domain

Note:

If you do not see the **TrailDuration** domain, try closing the **Domains** view and reopening it.

9. Using the **Table To Domain** geoprocessing tool, create four additional coded value domains using the parameters in the following table:

Input Table	Code Field	Description Field	Input Workspace	Domain Name	Domain Description
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Input Table	Code Field	Description Field	Input Workspace	Domain Name	Domain Description
Scenery	SceneryCode	SceneryDescription	vienna.gdb	Scenery	Various types of scenery along trail
TrailLength	LengthCode	LengthDescription	vienna.gdb	TrailLength	Trail distance in kilometers
ElevationChange	ElevationCode	ElevationDescription	vienna.gdb	ElevationChange	Rate of change in elevation along trail
TrailType	TrailTypeCode	TrailTypeDescription	vienna.gdb	TrailType	Description of trail types

10. Close and reopen the **Domains** view.

The **Scenery**, **TrailLength**, **ElevationChange**, and **TrailType** domains have been successfully added.

Domain Name	Description
Accessibility	Trail accessibility
ElevationChange	Rate of change in elevation along trail
LongTrailLength	Average kilometer length of a long trail
MaxTrailDuration	Maximum Trail Duration
MediumTrailLength	Average kilometer length of a medium trail
MinTrailDuration	Minimum Trail Duration
Scenery	Various types of scenery along trail
ShortTrailLength	Average kilometer length of a short trail
TrailDuration	Time needed to complete trail
TrailLength	Trail distance in kilometers
TrailType	Description of trail types

11. Close the **Domains** view and save the project.

Assign domains to fields

If the features in a table have been grouped into subtypes, different attribute domains can be assigned to each of the subtypes. However, when a domain is associated with an attribute field, only the values within that domain are valid for the field; the field will not accept a value that is not in that domain.

1. In the **Contents** pane, right-click the **Vienna City Trails** layer and choose **Attribute table**.

The **Trail Accessibility** and **Average Trail Elevation** fields are populated with non-descriptive values that are not very useful.

Trail Length	Trail Accessibility	Highlighted Scenery	Average Trail Elevation	Type of trail
11	2	1	2	< Null >
12	2	1	2	< Null >
2	2	1	2	< Null >
10	2	4	2	< Null >
11	2	5	2	< Null >
7	2	2	2	< Null >
10	2	1	2	< Null >
10	2	1	2	< Null >

You'll assign domains to these fields and review how domains affect the table.

- On the ribbon, on the **Data** tab, in the **Data Design** group, click **Subtypes**.
- If necessary, expand the **Subtypes** view so you can see the entire table without scrolling.
- For the **Easy** subtype, locate the **Accessibility** field. Double-click the **Domain** cell and choose **Accessibility**.

Subtype Name >		Easy	
Field Name	Data Type	Domain	Default Value
OBJECTID_1	Object ID		
Shape	Geometry		
OBJECTID	Double		
BEZ_TEXT	Text		
URL_INFO	Text		
TrailName	Text		
TrailStart	Text		
TrailEnd	Text		
Duration	Text		
Facilities	Text		
RouteDescription	Text		
MinDuration	Short		
MaxDuration	Short		
Length	Short		
Accessibility	Short	Accessibility ▾	
Scenery	Text		

- For the **Moderate**, **Strenuous**, and **Unrated** subtypes, assign the **Accessibility** domain to the **Accessibility** field.
- On the ribbon, on the **Subtypes** tab, in the **Changes** group, click **Save**.

Subtype Name >		Easy		Moderate		Strenuous		Unrated	
Field Name	Data Type	Domain	Default Value	Domain	Default Value	Domain	Default Value	Domain	Default Value
MinDuration	Short								
MaxDuration	Short								
Length	Short								
Accessibility	Short	Accessibility		Accessibility		Accessibility		Accessibility	
Scenery	Text								
Elevation	Short								

Note:

A benefit of applying a subtype to a field in a table is that you can apply different domains for each subtype for the same attribute field.

7. For the **Length** field, set the following domains:

- For the **Easy** subtype, set **Domain** to **ShortTrailLength**.
- For the **Moderate** subtype, set **Domain** to **MediumTrailLength**.
- For the **Strenuous** subtype, set **Domain** to **LongTrailLength**.
- For the **Unrated** subtype, set **Domain** to **ShortTrailLength**.

*Subtypes: Vienna City Trails									
Subtype Name >		Easy		Moderate		Strenuous		Unrated	
Field Name	Data Type	Domain	Default Value	Domain	Default Value	Domain	Default Value	Domain	Default Value
MinDuration	Short								
MaxDuration	Short								
Length	Short	ShortTrailLength		MediumTrailLength		LongTrailLength		ShortTrailLength	
Accessibility	Short	Accessibility		Accessibility		Accessibility		Accessibility	
Scenery	Text								

8. On the ribbon, on the **Subtypes** tab, in the **Changes** group, click **Save**.

9. For the **Easy**, **Moderate**, **Strenuous**, and **Unrated** subtypes, assign the following additional domains:

Field Name	Domain
MinDuration	MinTrailDuration
MaxDuration	MaxTrailDuration
Scenery	Scenery
Elevation	ElevationChange
TrailType	TrailType

*Subtypes: Vienna City Trails									
Subtype Name >		Easy		Moderate		Strenuous		Unrated	
Field Name	Data Type	Domain	Default Value	Domain	Default Value	Domain	Default Value	Domain	Default Value
RouteDescription	Text								
MinDuration	Short	MinTrailDuration		MinTrailDuration		MinTrailDuration		MinTrailDuration	
MaxDuration	Short	MaxTrailDuration		MaxTrailDuration		MaxTrailDuration		MaxTrailDuration	
Length	Short	ShortTrailLength		MediumTrailLength		LongTrailLength		ShortTrailLength	
Accessibility	Short	Accessibility		Accessibility		Accessibility		Accessibility	
Scenery	Text	Scenery		Scenery		Scenery		Scenery	
Elevation	Short	ElevationChange		ElevationChange		ElevationChange		ElevationChange	
TrailType	Text	TrailType		TrailType		TrailType		TrailType	
TrailCode	Short								

10. On the ribbon, on the **Subtypes** tab, in the **Changes** group, click **Save**.

11. Close the **Subtypes** view.

12. If necessary, open the attribute table for the **Vienna City Trails** layer.

The **Trail Accessibility**, **Highlighted Scenery**, and **Average Trail Elevation** fields are now populated with descriptive values derived from the domain descriptions.

Trail Accessibility	Highlighted Scenery	Average Trail Elevation	Type of trail
Paved	Forest	100 - 300 Meters	< Null >
Paved	Forest	100 - 300 Meters	< Null >
Paved	Forest	100 - 300 Meters	< Null >
Paved	Stream	100 - 300 Meters	< Null >
Paved	Bird watching	100 - 300 Meters	< Null >
Paved	River	100 - 300 Meters	< Null >

The **Type of trail** field remains empty, as no values have been entered into to this field. This field does have a domain applied, but without values added and with no default set, the domain cannot be validated.

13. Close the attribute table and save the project.

Update a trail feature

Next, you'll update a trail feature and explore the effect of domains while editing.

Using domains helps ensure data integrity by limiting the choice of values for a particular field. Validation for coded value domains is accomplished by restricting field values found in drop-down lists. Range domains are automatically validated during editing. You'll update a trail feature and use a code value and range domain.

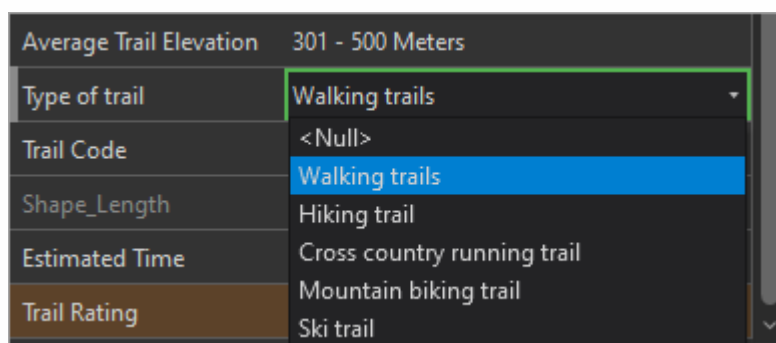
1. Click the **Source** tab to return to the original map view.



2. In the **Contents** pane, click the **Vienna City Trails** layer to select it.
3. On the ribbon, click the **Edit** tab. In the **Selections** group, click the **Select** button.
4. On the map, click any trail feature to select it.
5. On the **Edit** tab, in the **Selections** group, click **Attributes**.

The **Attributes** pane appears. You used this pane previously in the tutorial to edit field values for a selected trail. This time, the values you can enter and the choices you can apply to fields are affected by the domains applied to the fields.

6. In the **Attributes** pane, click the **Type of trail** value.



You cannot type your own trail type. Instead, you must choose one of the predefined types.

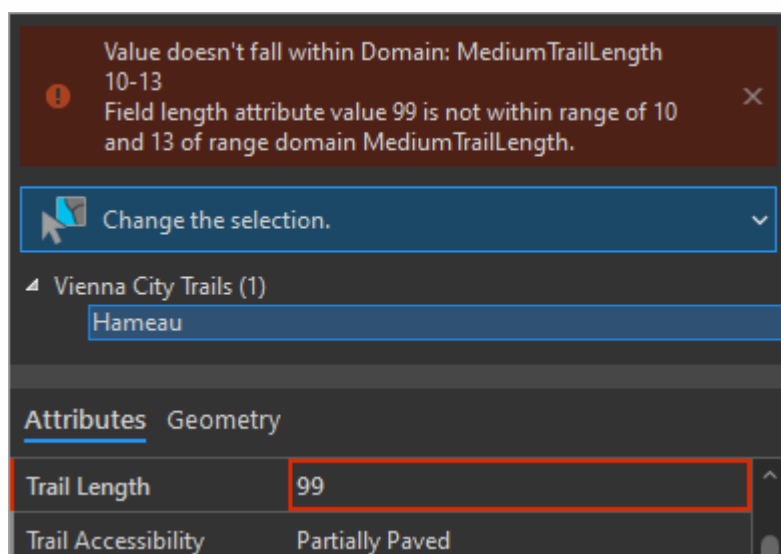
7. For **Type of trail**, choose any type that is different from the original trail type.
8. Experiment with changing the values for **Trail Accessibility** and **Highlighted Scenery**.

Previously, you assigned a range domain to the **Trail Length** field that restricts length based on the different subtypes as shown in the following table:

Subtype	Domain Name	Minimum Value	Maximum Value
Easy	ShortTrailLength	2	9
Moderate	MediumTrailLength	10	13
Strenuous	LongTrailLength	14	15

9. In the **Attributes** pane, for **Trail Length**, type 99 and press Enter.

The field highlights in red, indicating that the trail length is not within the range allowed for a moderate trail rating subtype. In addition, the **Attributes** pane displays a warning that the medium trail length allowed must be between 10 and 13.



10. Update the **Trail Length** value to 11 and press Enter.

The value falls within the allowable range and the warning message no longer appears.

Your current changes are not committed back to the geodatabase until you save your edits.

11. Close the **Attributes** pane.

A message appears, informing you that there are uncommitted attribute edits. You only made changes to experiment with editing, so you'll discard the changes.

12. Click **Discard**.
13. Clear the selection.

14. On the ribbon, click the **Map** tab. In the **Navigate** group, click the **Explore** button to deactivate the **Select** tool.
15. Save the project.

You've added domains and applied them to fields of a feature class. You also edited a feature with domains and subtypes. Next, you'll publish and use your feature service.

Publish a feature service

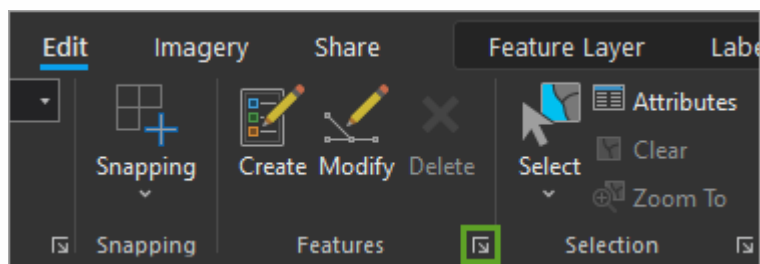
Next, you'll publish the **Vienna City Trails** layer and verify access to data validation while editing in a web map.

Share a web layer

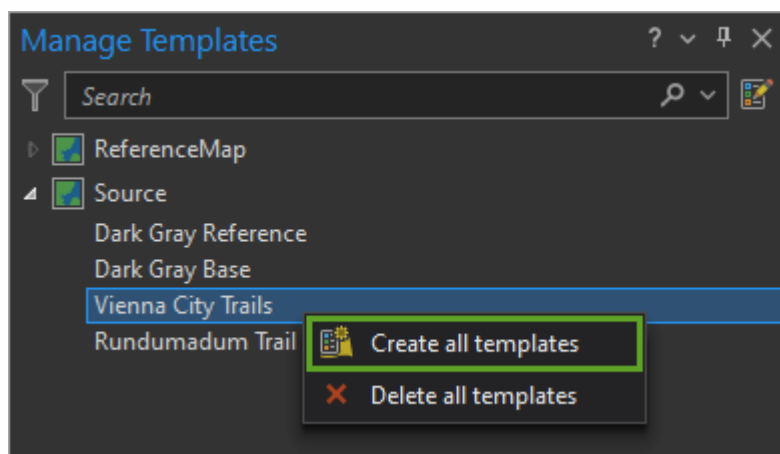
Before you can share features that support subtypes and domains in a web app, you must ensure that all editable layers have a feature template configured. A feature template is a predefined collection of construction tools that is used to create features on the layer.

Feature templates are a property of a layer and are managed and maintained in the ArcGIS Pro project. If the layer is shared as a layer file, layer package, or web layer, the template properties are also shared. You'll create feature templates for the **Vienna City Trails** layer before publishing the layer.

1. On the **Edit** tab, click the **Manage Templates** button in the lower corner of the **Features** group.



2. In the **Manage Templates** pane, right-click the **Vienna City Trails** layer and click **Create all templates**.



Feature templates are generated and added to the templates box for each associated layer subtype. Each template is named to match the subtype description. You can set different default values and apply different edit strategies and edit tools to be used to update or create features using the templates.

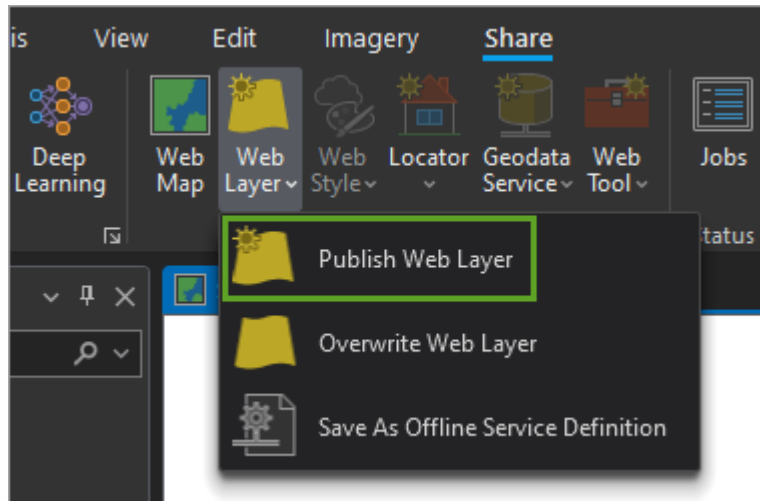
You're ready to publish the layer. First, you'll remove the other layers from the map and save the project.

3. In the **Contents** pane, remove all layers (even the basemap layers) except the **Vienna City Trails** layer.

Tip:

Remove a layer by right-clicking the layer and choosing **Remove**.

4. Save the project.
5. On the ribbon, click the **Share** tab. In the **Share As** group, click the **Web Layer** drop-down menu and choose **Publish Web Layer**.



The **Share As Web Layer** pane appears.

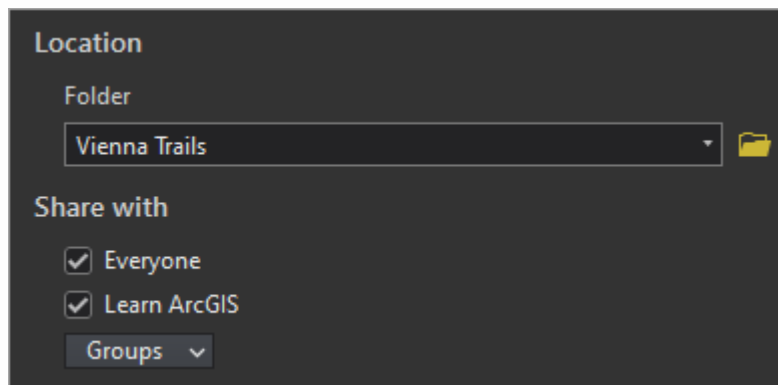
6. In the **Share As Web Layer** pane, update the following parameters:
- For **Name**, type Vienna City Trails followed by your name or initials.
 - For **Summary**, type Hiking trail data with attribute validation applied.
 - For **Tags**, keep the default list of tags.
 - For **Layer Type**, confirm that **Feature** is selected.

A screenshot of the 'Share As Web Layer' pane in ArcGIS. The pane has three tabs: 'General' (selected), 'Configuration', and 'Content'. Under the 'General' tab, there is a section titled 'Item Details'. It contains three input fields: 'Name' with the text 'Vienna City Trails Your Name', 'Summary' with the text 'Hiking trail data with attribute validation applied', and 'Tags' with a list of tags: 'Vienna', 'Austria', 'Visitor map', 'Tourism', 'Tourist', and 'Visitor guide'. Below the 'Item Details' section is a 'Layer Type' section with three radio buttons: 'Feature' (selected), 'Tile', and 'Vector Tile'. There is also a checkbox labeled 'Feature' which is checked.

Note:

If you're an ArcGIS Enterprise user, under **Data and Layer Type**, for **Copy all data**, choose **Feature**.

- For **Location**, type Vienna Trails. For **Share with**, check the box for **Everyone**.



Before you publish, you'll analyze the layer for any errors that would prevent publishing.

- In the **Share As Web Layer** pane, click **Analyze**.

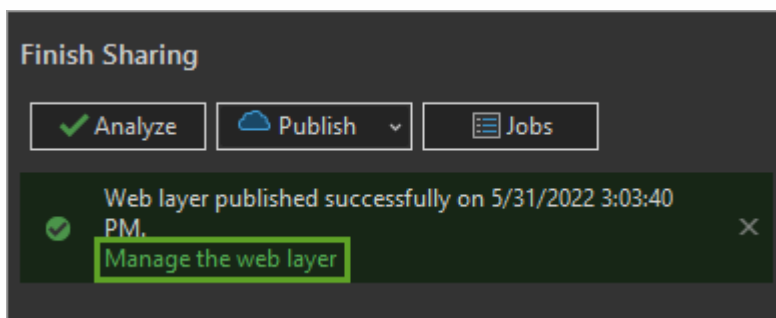
A warning may appear related to the assignment of unique numeric IDs for sharing web layers.

- If the warning appears, double-click the warning. In the **Map Properties** window, check **Allow assignment of unique numeric IDs for sharing web layers** and click **OK**.
- If necessary, in the **Share As Web Layer** pane, click **Analyze**. Confirm that there are no errors found.

Note:

If you encounter an error other than the unique numeric IDs error, read the error's description and resolve it before continuing.

- Click **Publish**.
- After publishing is finished, click **Manage the web layer**.



A browser opens to the item page for your Vienna City Trails web layer.

Enable editing

Next, you'll enable editing for your web layer.

- If necessary, on the item page, sign in to your ArcGIS organizational account.
- On the ribbon, click **Settings**.

- Under **General**, for **Public Data Collection**, check the box for **Approve this layer to be shared with the public when editing is enabled**.

General

Content Status

Discourage the use of this item.

Mark as Deprecated

Delete Protection

☐ Prevent this item from being accidentally deleted.

Public Data Collection

☒ Approve this layer to be shared with the public when editing is enabled.

- Next to **General**, click **Save**.
- Under **Feature Layer (hosted)**, for **Editing**, check **Enable editing**.

Feature Layer (hosted)

Editing

☒ Enable editing

☐ Keep track of changes to the data (add, update, delete features).

☐ Keep track of who edited the data (editor name, date and time).

☐ Enable Sync (required for offline use and collaboration).

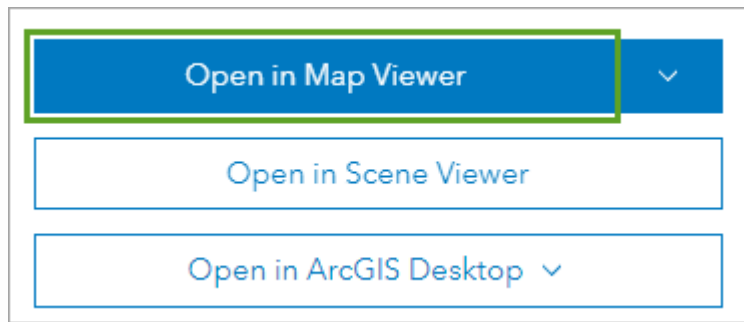
- Next to **Feature Layer (hosted)**, click **Save**.

Edit features in a web map

Now that editing is enabled for your layer, you'll add it to a web map and edit some features.

- On the ribbon, click **Overview**.

2. Click **Open in Map Viewer**.



The map opens in Map Viewer.

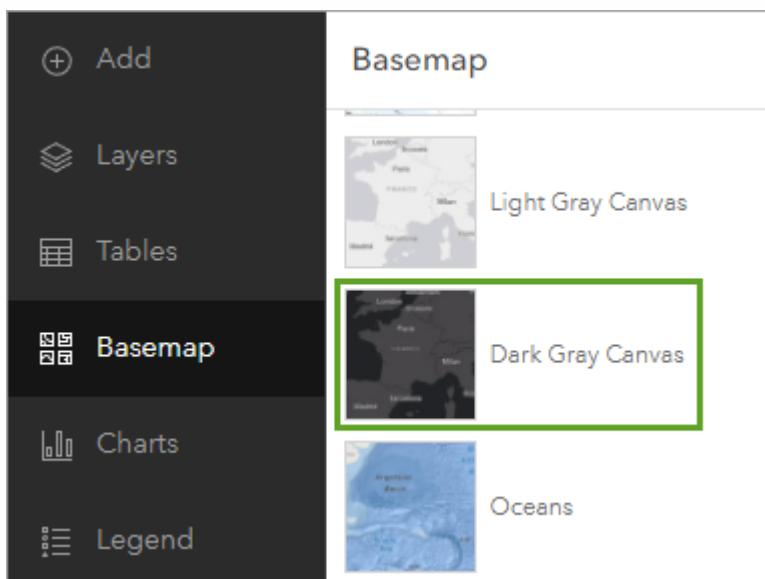
Note:

Depending on your organizational and user settings, you may have opened Map Viewer Classic. ArcGIS Online offers two map viewers for viewing, using, and creating maps. For more information on the map viewers available and which to use, please see this [FAQ](#).

3. If necessary, in the pop-up window, click **Open in Map Viewer** or on the ribbon, click **Open in Map Viewer**.

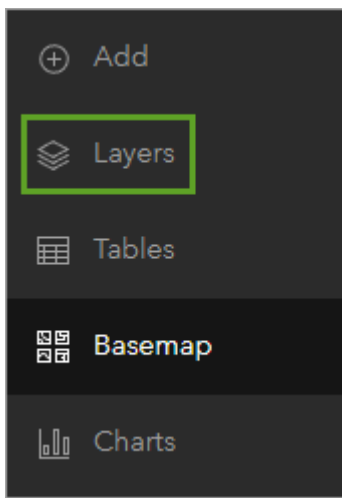
Map Viewer opens.

4. On the **Contents** (dark) toolbar, click **Basemap**. In the **Basemap** pane, choose **Dark Gray Canvas**.



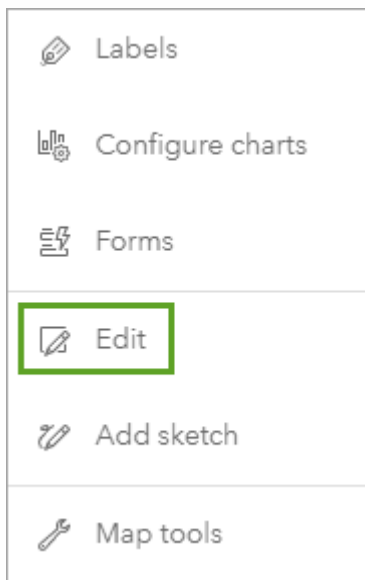
The basemap updates on your map.

5. On the **Contents** toolbar, click **Layers**.



The **Layers** pane appears.

- On the **Layers** pane, confirm that the **Vienna City Trails** layer is selected. On the **Settings** (light) toolbar, click **Edit**.



Tip:

If the **Edit** tool is not visible, click **More** to see more tools on the **Settings** toolbar and click **Edit**.

The **Editor** pane appears.

- In the **Editor** pane, under **Edit features**, click **Select**.

Editor

 Snapping

Edit features

 Select

Create features

Filter types

8. Click any trail on the map.

The trail is selected. The **Editor** pane changes to the **Edit feature** pane, showing all the attributes, subtypes, and domains that have been applied to the source layer, ensuring that new trails collected and edits made to trails in a web map or app will remain valid.

9. In the **Edit feature** pane, for **Type of trail**, choose **Hiking trail**.

Average Trail Elevation

100 - 300 Meters

Type of trail

Hiking trail

Trail Code

2

10. Click **Trail Length**, type 99, and press Enter.

A warning informs you that the value you entered is out of the minimum and maximum range as defined for this field.

Maximum Walk Time

4

Trail Length

99

 Value should be between 10 and 13

Trail Accessibility

Paved

Some other fields may have warnings as well. These are a result of changes made before the domain was applied to the field, and the value stored is now out of range.

11. For **Trail Length**, type 10 and press Enter.

Note:

The edits you make to the feature layer in Map Viewer are automatically saved to the layer. When you add, delete, or edit a feature or attribute and realize you made a mistake, you can click **Delete** to delete your edit.

12. At the bottom of the **Edit feature** pane, click **Update**.

The pane changes to the **Select feature** pane. You'll return to the **Editor** pane and create a new trail feature.

13. In the **Select feature** pane, click the back button. In the **Editor** pane, under **Create features**, click **Moderate** to add a new feature.

Create features

Filter types

Vienna City Trails Your Name

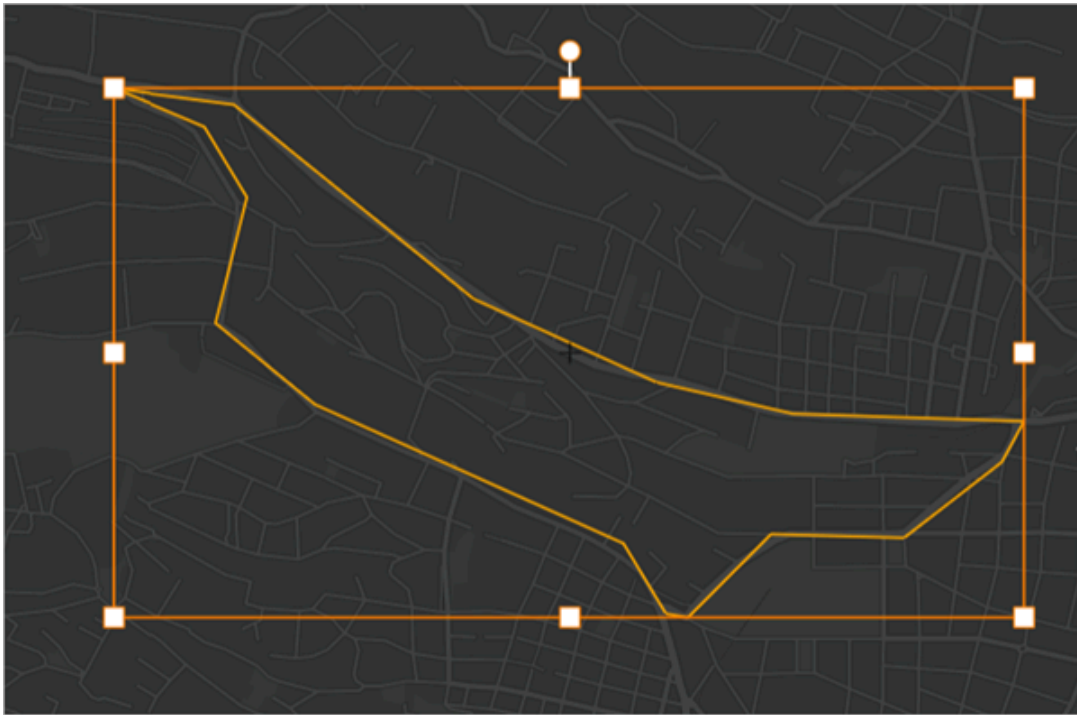
☒ Easy

☒ Moderate

☐ Strenuous

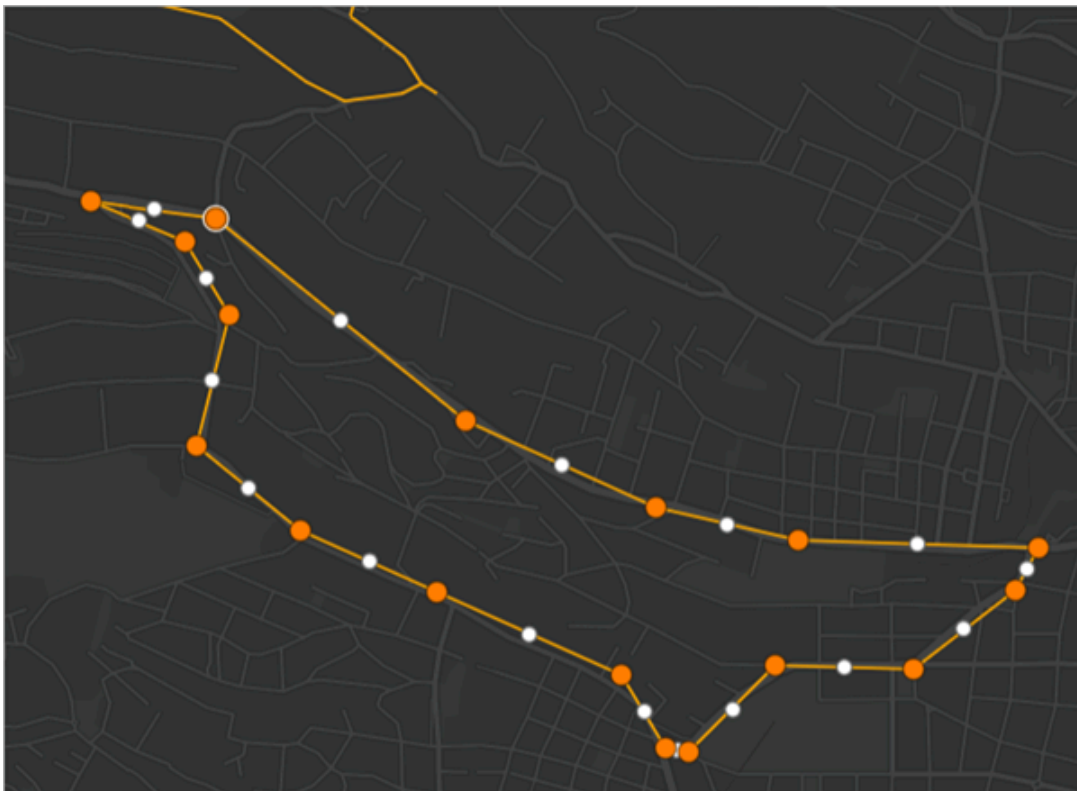
☐ Unrated

14. On the map, click anywhere to start adding vertices for a new trail feature and double-click when you complete the new trail.



15. Click the new feature to display its vertices.

You can click and move vertices as needed to update and change the route the trail follows.



16. To commit your updates, click a location on the map away from the feature you created.

The **Create features** pane displays the attributes for the new trail you created.

17. In the **Create features** window, update your new trail attributes by typing a name and selecting attribute values from the various lists applied to fields. When you're finished, click **Create**.

18. Optionally, save the web map.

In this tutorial, you investigated how a local Vienna tourist agency can use attribute validation techniques in a geodatabase to build and maintain a hiking trail layer that provides clients with trail information and offers them an opportunity to contribute and suggest new trails. Using subtypes and domains, the agency can ensure data consistency, reduce redundancy, and build efficient web maps and downloadable apps.

Subtypes and domains are powerful tools for ensuring data integrity and the efficiency of a geodatabase, and you may find them useful on many of your datasets. However, they are most useful when working with large datasets. Errors can occur in any dataset, but once your feature count exceeds a few hundred, it becomes difficult to manually sort through and ensure that you don't have any typos or other errors. If multiple people will be editing the same feature class, it increases the opportunity for errors.

Assigning domains to attribute fields creates restrictions on the allowable values, so you can't enter misspelled words or numeric values outside of a defined range. Domains help to ensure that your data remains clean and accurate. Subtypes can also apply restrictions on allowable attribute values but only on one integer field per feature class. Subtypes true usefulness is classifying your data to make it perform faster by storing information in coded integers instead of long strings.

On your own, consider revisiting the **Rundumadum Trail Stages** layer and assign geodatabase domains to the various attribute fields associated with the 24 stages of the trail.

You can find more tutorials in the [tutorial gallery](#).

Acknowledgements

- The ViennaCityTrails_Raw and RundumadumTrail_Subtypes feature classes are derived from the [Wanderwege](#) shapefile published as Open Government Data (OGD) by the City of Vienna (Data source: City of Vienna - [data.wien.gv.at](#)). The feature classes have additional attribute information derived from hiking trail information on the [City of Vienna website](#).
- The Vienna City Map shown in this tutorial contains data from the Austrian OGD basemap (Data source: [basemap.at](#)).
- The Duration, ElevationChange, Scenery, TrailLength, and TrailType tables were created by the author.
- Gallery card image by Brian Erickson on [Unsplash](#).
- Banner image by Simon Berger on [Pexels](#).

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